ON THE REPRESENTATION OF FUNCTIONS AS FOURIER TRANSFORMS OF UNBOUNDED MEASURES

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1. Introduction

In this article, we show that it is possible to assign to every member f in a certain class $R(\Phi)$ of functions on $(-\infty, \infty)$ a 'measure' μ such that

$$\sqrt{(2\pi)}f(x) = (\mathrm{C}, 1)\int e^{-ixt}\,d\mu(t), \tag{1}$$

almost everywhere; moreover, if f is locally in L^p for some p, $1 \leqslant p \leqslant \infty$, then

$$\left[\sum_{-\infty}^{\infty} [|\mu|([n, n+1))]^{p'}\right]^{1/p'} < \infty,$$

where $p' = p/(p-1)$; and if f is bounded below, then f is bounded on the average, in the sense that

$$\lim_{T\to\infty} \frac{1}{2T}\int_{-T}^{T} f(t)\,dt$$

exists and is finite.

The motivation for the present work stems from Cooper's paper ([**3**]), where he generalized the notion of a positive definite function, which owes its inception to Mathias ([**7**]) and its importance to Bochner ([**1**]). It will turn out that every locally summable function that is positive definite in Cooper's sense belongs to $R(\Phi)$; such a function admits a representation of type (1) in terms of a *positive* measure.

The symbol (C, 1) prefixing the integral appearing in (1) means that the integral is summable by Cesàro's method of order 1 to the value $\sqrt{(2\pi)}f(x)$, and we adhere to this notation throughout the paper. Other methods of summability could also be used: in particular, Cesàro's method of order α, for every positive α, and Abel's method; and it can be shown that the integral (1) is summable by any of these methods to $\sqrt{(2\pi)}f(x)$.

2. Notation and terminology

For any class J of functions defined on $(-\infty, \infty)$, we shall write J_c for the set of elements in J that have compact support, J_0 for the collection

of members of J that vanish at infinity and J_{loc} for the functions on $(-\infty,\infty)$ whose restrictions to compact sets are in J.

Besides the familiar L^p-spaces and the space C of continuous functions on $(-\infty,\infty)$, we shall introduce the Banach spaces (L^p,l^q) and (C_0,l^q); these are defined as follows.

When $1 \leqslant p,\ q \leqslant \infty$, (L^p,l^q) denotes the class of functions $f \in L^p{}_{\text{loc}}$ with finite norm

$$\|f\|_{p,q} = \left[\sum_{-\infty}^{\infty}\left[\int_n^{n+1}|f(x)|^p\,dx\right]^{q/p}\right]^{1/q};$$

and (C_0,l^q) stands for $(L^\infty,l^q)\cap C_0$, with

$$\|f\|_{\infty,q} = \left[\sum_{-\infty}^{\infty}\max_{n\leqslant x\leqslant n+1}|f(x)|^q\right]^{1/q}$$

as norm.

We shall use the term 'measure' to mean a set function μ that takes complex values on compact subsets of $(-\infty,\infty)$ and is such that, for every compact set K, the set function $E \mapsto \mu(E\cap K)$ is a complex measure on the Borel subsets of $(-\infty,\infty)$. If μ is such a measure and for some r, $1 \leqslant r \leqslant \infty$,

$$\left[\sum_{-\infty}^{\infty}[|\mu|([n,n+1))]^r\right]^{1/r} \tag{2}$$

is finite, we say that $\mu \in M_r$, and write $\|\mu\|_{M_r}$ for the sum of the series (2).

We shall write $\hat{f}$ and $g*h$ for the Fourier transform of f and the convolution of g and h, respectively. Thus

$$\sqrt{(2\pi)}\hat{f}(x) = \int e^{-ixt}f(t)\,dt,$$

and

$$(g*h)(x) = \int g(x-t)h(t)\,dt.$$

Fejér's kernel will be denoted by $\{k_\lambda : \lambda > 0\}$, that is,

$$k_\lambda(x) = \frac{1-\cos\lambda x}{\pi\lambda x^2},$$

so that

$$\sqrt{(2\pi)}\hat{k}_\lambda(x) = \max(1-|x/\lambda|,0)$$

and the Fourier transform of $\hat{k}_\lambda$ is k_λ.

3. Background material

In this section, we summarize for easy reference some properties of the spaces (L^p,l^q), (C_0,l^q), and M_r; full proofs of Theorems A, B, C, and D can be found in [6].

THEOREM A. *If* $1 \leqslant p, q \leqslant \infty$, *then the Banach dual of* (L^p, l^q) *is (isometrically isomorphic to)* $(L^{p'}, l^{q'})$.

THEOREM B. *Let* $1 \leqslant q \leqslant \infty$. *If* T *is a continuous linear functional on* (C_0, l^q), *then there is a measure* $\mu \in M_{q'}$ *such that*

$$T(f) = \int f d\mu \quad (f \in (C_0, l^q));$$

moreover,

$$\|T\| \leqslant \|\mu\|_{M_{q'}} \leqslant 2^{1/q}\|T\|.$$

THEOREM C. *If* $1 \leqslant p \leqslant 2$, *then the Fourier transform of an element of* (L^p, l) *belongs to* $(C_0, l^{p'})$ *and there is a constant* A_p *such that*

$$\|\hat{f}\|_{\infty,p'} \leqslant A_p \|f\|_{p,1} \quad (f \in (L^p, l)).$$

THEOREM D. *Let* $1 \leqslant q \leqslant 2$ *and let* $\mu \in M_q$. *Then, as* $N \to \infty$,

$$\frac{1}{\sqrt{(2\pi)}} \int_{-N}^{N} e^{-ixt} d\mu(t)$$

converges in the norm of $(L^{q'}, l^\infty)$ *to a function* $\hat{\mu}$ *and*

$$\int h(x)\hat{\mu}(x)\,dx = \int \hat{h}\, d\mu \quad (h \in (L^q, l)).$$

Further,

$$\sqrt{(2\pi)}\hat{\mu}(x) = (\mathrm{C}, 1)\int e^{-ixt} d\mu(t),$$

almost everywhere.

We shall also require the following result, the proof of which is straightforward and will be omitted.

THEOREM E. *If* $1 \leqslant q \leqslant \infty$ *and* $f \in (C_0, l^q)$, *then*

$$\lim_{\lambda \to \infty} \|k_\lambda * f - f\|_{\infty,q} = 0.$$

4. Classes of test functions

Let $1 \leqslant q \leqslant \infty$. The vector space of functions

$$\{\varphi \in C_c : \hat{\varphi} \in (C_0, l^q)\}, \quad \text{with} \quad \varphi \mapsto \|\hat{\varphi}\|_{\infty,q}$$

as norm, will be denoted by Φ_q. When $q = 1$, we write Φ in place of Φ_1.

Clearly, none of these spaces contains only the zero function; and the inclusion relations

$$\Phi \subseteq \Phi_q \subseteq \Phi_\infty \quad (1 \leqslant q \leqslant \infty)$$

are obvious.

If $2 \leqslant q \leqslant \infty$ and $\psi \in C_c$, then by Theorem C, $\hat{\psi} \in (C_0, l^q)$. Hence, if we equip C_c with the norm of Φ_q, in this case, we can identify Φ_q with C_c.

On the other hand, if $1 \leqslant q < 2$, then not every function in C_c belongs to Φ_q. To see this, consider the function

$$\theta(x) = \begin{cases} \sum_{n=2}^{\infty} \dfrac{e^{in \log n} \sin nx}{n^{\frac{1}{2}}(\log n)^{\beta}} & \text{if } |x| \leqslant \pi, \\ 0 & \text{otherwise,} \end{cases}$$

where $\beta > 1$. Since the series

$$\sum_{n=2}^{\infty} \frac{e^{icn \log n} e^{inx}}{n^{\frac{1}{2}}(\log n)^{\beta}}$$

converges uniformly on $[-\pi, \pi]$ for every real c ([13], p. 199), $\theta \in C_c$; and since

$$\hat{\theta}(n) = \sqrt{(\pi/2)} \frac{e^{in \log n}}{n^{\frac{1}{2}}(\log n)^{\beta}} \quad (|n| \geqslant 2),$$

$$\sum_{-\infty}^{\infty} |\hat{\theta}(n)|^q = \infty,$$

whenever $1 \leqslant q < 2$. Hence $\theta \notin \Phi_q$ for this range of q.

5. Resonance classes

Let $1 \leqslant q \leqslant \infty$. A measurable function f on $(-\infty, \infty)$ is said to be *resonant relative to the test space* Φ_q if

(i) $f\varphi$ is Lebesgue-summable for all $\varphi \in \Phi_q$, and

(ii) the linear functional $\varphi \mapsto \int f(x)\varphi(x)\,dx$ is continuous on Φ_q.

The collection of functions resonant relative to Φ_q will be designated by $R(\Phi_q)$.

The following statements are more or less immediate consequences of the definition and we omit the easy proofs.

(a) $R(\Phi_\infty) \subseteq R(\Phi_q) \subseteq R(\Phi)$ $(1 \leqslant q \leqslant \infty)$.

(b) Each $R(\Phi_q)$ is a vector space.

(c) If $f \in R(\Phi_q)$, then $\tilde{f} \in R(\Phi_q)$, where $\tilde{f}(x) = \overline{f(-x)}$.

(d) If $f \in R(\Phi_q)$, then $x \mapsto e^{ixt} f(x) \in R(\Phi_q)$, for every real t.

(e) If $f \in R(\Phi_q)$, then $x \mapsto f(x-t) \in R(\Phi_q)$, for every real t.

We proceed to establish some integrability conditions on the elements of $R(\Phi_q)$.

THEOREM 1. *If f is resonant relative to Φ, then f is locally summable on $(-\infty, \infty)$.*

Proof. Choose a compact set K and select a function φ_K in C_c so that $\varphi_K(x) = 1$ on K and $\hat{\varphi}_K \in (C_0, l)$. By hypothesis, $f\varphi_K \in L$. Since

$|f(x)| \leqslant |f(x)\varphi_K(x)|$ for all $x \in K$, we deduce that f is integrable on K. Thus $f \in L_{\text{loc}}$, as claimed.

THEOREM 2. *Let $2 \leqslant q \leqslant \infty$ and $f \in R(\Phi_q)$. Then $f \in (L^q, l^\infty)$.*

Proof. By hypothesis, $\varphi \mapsto \int f(x)\varphi(x)\,dx$ is continuous on Φ_q; and $C_c \subseteq \Phi_q$. Hence there is a positive constant A such that

$$\left|\int f(x)\varphi(x)\,dx\right| \leqslant A\,\|\hat{\varphi}\|_{\infty,q} \quad (\varphi \in C_c).$$

Now apply Theorem C. Then there is a constant B_q such that

$$\left|\int f(x)\varphi(x)\,dx\right| \leqslant B_q\,\|\varphi\|_{q',1} \quad (\varphi \in C_c).$$

Consequently, the linear functional $\varphi \mapsto \int f(x)\varphi(x)\,dx$ is continuous on a dense subspace of $(L^{q'}, l)$ and so, by Theorem A, $f \in (L^q, l^\infty)$, as required.

In view of these two results, it is tempting to conjecture that if $f \in R(\Phi_q)$, $1 \leqslant q < 2$, then $f \in L^q_{\text{loc}}$; but we are unable to decide the matter.

The next theorem shows that if $1 < q < 2$, $f \in R(\Phi_q)$, and if, moreover, $f \in L^q_{\text{loc}}$, then f is actually resonant relative to a larger class of functions.

THEOREM 3. *Let $1 < q < 2$ and assume that f is resonant relative to Φ_q. Assume, in addition, that $|f|^q$ is locally summable on $(-\infty, \infty)$. Then there is a constant A, depending only on q, such that*

$$\left|\int f(x)\psi(x)\,dx\right| \leqslant A\,\|\hat{\psi}\|_{\infty,q}$$

for all $\psi \in L^q_c$ such that $\hat{\psi} \in (C_0, l^q)$.

Proof. At any rate, there is a constant A (depending on q) such that

$$\left|\int f(x)\varphi(x)\,dx\right| \leqslant A\,\|\hat{\varphi}\|_{\infty,q} \tag{3}$$

for all $\varphi \in \Phi_q$.

Let ψ be a function in $L^{q'}_c$ such that $\hat{\psi} \in (C_0, l^q)$ and consider the family of functions $\{\psi_h : h > 0\}$ defined by

$$\psi_h(x) = \frac{1}{2h}\int_{-h}^{h} \psi(x+t)\,dt \quad (-\infty < x < \infty).$$

For each $h > 0$, $\psi_h \in C_c$; and $\|\psi_h - \psi\|_{q'} \to 0$ as $h \to 0$. Also

$$\hat{\psi}_h(x) = \hat{\psi}(x)\frac{\sin hx}{hx},$$

whence it follows that

$$\|\hat{\psi}_h\|_{\infty,q} \leqslant \|\hat{\psi}\|_{\infty,q}. \tag{4}$$

Hence $\{\psi_h : h > 0\}$ is a subset of Φ_q and so, by (3) and (4),

$$\left|\int f(x)\psi_h(x)\,dx\right| \leqslant A\,\|\hat{\psi}\|_{\infty,q}.$$

The conclusion now results from this on making $h \to 0$.

REMARK. The theorem is also true when $q = 1$.

6. Other conditions for resonance

THEOREM 4. *Let $f \in R(\Phi_q)$, where $1 \leqslant q < \infty$. Let $g \in L$. Then $f\hat{g} \in R(\Phi_q)$.*

Proof. Let $\varphi \in \Phi_q$. Then $\psi = \varphi\hat{g}$ belongs to C_c and

$$\begin{aligned}\sqrt{(2\pi)}\hat{\psi}(x) &= \int e^{-ixt}\varphi(t)\hat{g}(t)\,dt \\ &= \int \hat{\varphi}(x+t)g(t)\,dt.\end{aligned}$$

Since, by Hölder's inequality,

$$(2\pi)^{q/2}\,|\hat{\psi}(x)|^q \leqslant \int |\hat{\varphi}(x+t)|^q\,|g(t)|\,dt.\|g\|_1^{q/q'},$$

we see that

$$\begin{aligned}(2\pi)^{q/2}\|\hat{\psi}\|_{\infty,q}^q &\leqslant \sum_{-\infty}^{\infty} \max_{n\leqslant x\leqslant n+1} \int |\hat{\varphi}(x+t)|^q\,|g(t)|\,dt.\|g\|_1^{q/q'} \\ &\leqslant 2\,\|\hat{\varphi}\|_{\infty,q}^q\,\|g\|_1\,\|g\|_1^{q/q'}.\end{aligned}$$

Hence $\psi \in \Phi_q$ and

$$\|\psi\|_{\Phi_q} \leqslant \frac{2^{1/q}\,\|g\|_1}{\sqrt{(2\pi)}}\,\|\varphi\|_{\Phi_q}.$$

Thus the linear operator $\varphi \mapsto \varphi\hat{g}$ is continuous on Φ_q. Consequently, the linear functional $\varphi \mapsto \int f(x)\hat{g}(x)\varphi(x)\,dx$ is continuous on Φ_q. In other words, $f\hat{g} \in R(\Phi_q)$, as we wanted to show.

The next theorem is a converse of the preceding one.

THEOREM 5. *Assume that $f\hat{g} \in R(\Phi_q)$ for all $g \in L$. Then $f \in R(\Phi_q)$.*

Proof. Evidently, f is locally summable and for each $g \in L$, $\varphi \mapsto \int f(x)\hat{g}(x)\varphi(x)\,dx$ is continuous on Φ_q.

Let $S = \{\varphi \in \Phi_q : \|\varphi\|_{\Phi_q} = 1\}$ and consider the family of linear functionals $\{T_\varphi : \varphi \in S\}$ defined on L by

$$T_\varphi(g) = \int f(x)\hat{g}(x)\varphi(x)\,dx \quad (g \in L).$$

Clearly, each T_φ is continuous on L; and, under the given conditions,

$$\sup\{|T_\varphi(g)| : \varphi \in S\} < \infty$$

for each $g \in L$. Hence, by the principle of uniform boundedness,

$$\sup\{\|T_\varphi\| : \varphi \in S\} < \infty,$$

that is, there is a constant A (depending only on q) such that

$$\left|\int f(x)\hat{g}(x)\varphi(x)\,dx\right| \leqslant A\,\|g\|_1,$$

whenever $\varphi \in S$ and $g \in L$. In particular, putting g equal to Fejér's kernel we deduce that

$$\left|\int f(x)\hat{k}_\lambda(x)\varphi(x)\,dx\right| \leqslant A\,\|k_\lambda\|_1 = A, \tag{5}$$

for all $\varphi \in S$ and all $\lambda > 0$. Now, $\sqrt{(2\pi)}\hat{k}_\lambda \to 1$ boundedly as $\lambda \to \infty$. In view of this and Lebesgue's dominated convergence theorem, it is permissible to make λ tend to infinity in (5). We infer that

$$\left|\int f(x)\varphi(x)\,dx\right| \leqslant A,$$

if only $\varphi \in S$. Hence $f \in R(\Phi_q)$.

This ends the proof.

7. Characterization of the classes $R(\Phi_q)$, $2 \leqslant q \leqslant \infty$

The problem of classifying those bounded continuous functions on $(-\infty, \infty)$ which are expressible as Fourier transforms of finite complex measures was solved by Bochner ([2]); an identical solution was discovered independently and almost simultaneously by Verblunsky ([11]). By modifying the Bochner–Verblunsky condition, Schoenberg ([9]) established a criterion for a function $f \in L^\infty$ to coincide almost everywhere with the Fourier transform of a finite complex measure. In our terminology, f admits such a representation if and only if $f \in R(\Phi_\infty)$. We proceed to describe the elements of $R(\Phi_q)$ for the remaining values of $q \geqslant 2$.

THEOREM 6. *Let $2 \leqslant q \leqslant \infty$. Then $f \in R(\Phi_q)$ if and only if there is a measure $\mu \in M_{q'}$ such that $f = \hat{\mu}$.*

Proof. Necessity. Let $f = \hat{\mu}$, where $\mu \in M_{q'}$. By Theorem D, $f \in (L^q, l^\infty)$ and

$$\int f(x)h(x)\,dx = \int \hat{h}\,d\mu$$

for all $h \in (L^{q'}, l)$. Hence

$$\left| \int f(x)h(x)\,dx \right| \leqslant \|\hat{h}\|_{\infty,q} \|\mu\|_{M_{q'}}$$

for all $h \in (L^{q'}, l)$. Consequently, $f \in R(\Phi_q)$.

Sufficiency. Suppose that $f \in R(\Phi_q)$. Then there is a constant A (depending on q) such that

$$\left| \int f(x)\varphi(x)\,dx \right| \leqslant A \|\hat{\varphi}\|_{\infty,q}$$

for all $\varphi \in C_c$. In other words, the linear functional $\hat{\varphi} \mapsto \int f(x)\varphi(x)\,dx$ is bounded on a subspace of (C_0, l^q) and so, invoking, in turn, the Hahn–Banach theorem and Theorem B, we infer the existence of an element $\mu \in M_{q'}$ such that

$$\int f(x)\varphi(x)\,dx = \int \hat{\varphi}\,d\mu \quad (\varphi \in C_c). \tag{6}$$

Since $f \in (L^q, l^\infty)$, by Theorem 2, every $\varphi \in (L^{q'}, l)$ satisfies (6). Appealing once more to Theorem D, we see that $f = \hat{\mu}$.

This ends the proof.

8. Concerning the classes $R(\Phi_q)$, $1 \leqslant q < 2$

THEOREM 7. *Let $1 \leqslant q < 2$. Suppose that f is resonant relative to Φ_q and that $f \in L^q_{\mathrm{loc}}$. Then there is a unique measure $\mu \in M_{q'}$ such that*

$$(\mathrm{C}, 1) \int \hat{h}(-x)f(x)\,dx = \int h\,d\mu \quad (h \in (C_0, l^q)). \tag{7}$$

Furthermore, if $p = 2q/(2q-1)$, then

$$\int\int f(x-y)\varphi(x)\overline{\psi(y)}\,dx\,dy = \sqrt{(2\pi)} \int \hat{\varphi}\overline{\hat{\psi}}\,d\mu \tag{8}$$

for all $\varphi, \psi \in (L^p, l)$. The double integral exists not necessarily as a Lebesgue integral, but as a limit of Lebesgue integrals over rectangles.

Proof. In virtue of Theorem 3 and the subsequent remark, there is a constant A (depending on q) such that

$$\left| \int f(x)\,\varphi(x)\,dx \right| \leqslant A \|\hat{\varphi}\|_{\infty,q}$$

whenever $\varphi \in L^{q'}_c$ and $\hat{\varphi} \in (C_0, l^q)$.

Manifestly, then, a repetition of the argument used in the sufficiency part of the previous theorem establishes the existence of a measure $\mu \in M_{q'}$ such that

$$\int f(x)\varphi(x)\,dx = \int \hat{\varphi}\,d\mu \tag{9}$$

whenever $\varphi \in L_c^q$ and $\hat{\varphi} \in (C_0, l^q)$. We proceed to show that μ is unique and that (7) holds. To do this, we consider the family of measures $\{\mu_\lambda : \lambda > 0\}$ given by

$$\mu_\lambda(E) = \int_E \left(\int \hat{k}_\lambda(u) f(u) e^{ixu} du \right) dx.$$

Since $\hat{k}_\lambda f \in L^q$, it is clear that $\mu_\lambda \in M_{q'}$. Next, if $h \in (C_0, l^q)$, $\hat{h} \in L^{q'}$ and

$$\begin{aligned} \int h \, d\mu_\lambda &= \int h(x) \left(\int \hat{k}_\lambda(u) f(u) \, e^{ixu} du \right) dx \\ &= \int \sqrt{(2\pi)} \hat{k}_\lambda(u) \hat{h}(-u) f(u) \, du \\ &= \int k_\lambda * h \, d\mu \\ &\to \int h \, d\mu \end{aligned}$$

as $\lambda \to \infty$. The transition from the first to the second line is justified by Parseval's theorem since $\hat{k}_\lambda f \in L^q$; the next step is a consequence of (9) since $\sqrt{(2\pi)} \hat{k}_\lambda(u) \hat{h}(-u) \in L_c^{q'}$ and its transform is $k_\lambda * h$; finally, Theorem E validates the limit relation.

Clearly, then, μ is unique and (7) holds.

Next, we prove (8) for functions in L_c^p. Let $\varphi, \psi \in L_c^p$. Noting that $1 < p \leqslant 2$ and that $p' = 2q$, it follows easily from Theorem C that $\hat{\varphi}\overline{\hat{\psi}} \in (C_0, l^q)$. Moreover, the Fourier transform of $\hat{\varphi}\overline{\hat{\psi}}$ is given by the convolution

$$\frac{1}{\sqrt{(2\pi)}} \int \overline{\psi(t)} \varphi(x+t) \, dt,$$

which belongs to $L_c^{q'}$. Hence, by (7) and Fubini's theorem,

$$\int\int f(x-y) \varphi(x) \overline{\psi(y)} \, dx \, dy = \sqrt{(2\pi)} \int \hat{\varphi}\overline{\hat{\psi}} \, d\mu. \tag{10}$$

Put

$$B(\varphi, \psi) = \int\int f(x-y) \varphi(x) \overline{\psi(y)} \, dx \, dy \quad (\varphi, \psi \in L_c^p).$$

Then, for all $\varphi, \psi \in L_c^p$,

$$\begin{aligned} |B(\varphi, \psi)| &\leqslant \sqrt{(2\pi)} \, \| \hat{\varphi}\hat{\psi} \|_{\infty, q} \| \mu \|_{M_{q'}} \\ &\leqslant \sqrt{(2\pi)} \, \| \hat{\varphi} \|_{\infty, p'} \| \hat{\psi} \|_{\infty, p'} \| \mu \|_{M_{q'}} \\ &\leqslant \sqrt{(2\pi)} A_p{}^2 \| \varphi \|_{p,1} \| \psi \|_{p,1} \| \mu \|_{M_{q'}} \end{aligned}$$

by Theorem C. If θ is any element in (L^p, l), we can write $\theta = \sum_{-\infty}^{\infty} \theta_n$, where θ_n vanishes outside $[n, n+1]$ and $\| \theta \|_{p,1} = \sum_{-\infty}^{\infty} \| \theta_n \|_{p,1}$; and using

this notation, we see that, for any pair of elements φ, ψ in (L^p, l),

$$\sum |B(\varphi_n, \psi_m)| \leqslant \sqrt{(2\pi)} A_p{}^2 \|\varphi\|_{p,1} \|\psi\|_{p,1}.$$

Thus the doubly infinite double series

$$\sum B(\varphi_n, \psi_m)$$

is absolutely convergent; that is, the integral on the left-hand side of (8) exists as a limit of Lebesgue integrals over rectangles, the coordinates of whose vertices are integers. Moreover, returning to (10) we see that the sum of the series is equal to

$$\sqrt{(2\pi)} \sum \int \hat{\varphi}_n \overline{\hat{\psi}}_m \, d\mu = \sqrt{(2\pi)} \int \hat{\varphi} \overline{\hat{\psi}} \, d\mu,$$

since $\sum \hat{\varphi}_n$ converges to $\hat{\varphi}$ in the norm of (C_0, l^p); and a similar remark applies to $\sum \hat{\psi}_n$. Thus (8) holds.

This completes the proof.

COROLLARY. *If* $1 < q < 2$, *then under the conditions of Theorem* 7,

$$\sqrt{(2\pi)} f(x) = \frac{d}{dx} \int \frac{e^{-ixt} - 1}{-it} \, d\mu(t),$$

almost everywhere.

Proof. This follows in a straightforward manner from (7).

9. Concerning the class $R(\Phi)$

Throughout the remainder of the paper we confine our attention to the class of functions that are resonant relative to Φ.

THEOREM 8. *Let* $f \in R(\Phi)$ *and assume that* f *is bounded below. Let* μ *be the measure in* M_∞ *associated with* f *by the previous theorem. Then*

(i) $f \in (L, l^\infty)$,

(ii) $\sqrt{(2\pi)} f(x) = (\mathrm{C}, 1) \int e^{-ixt} \, d\mu(t)$, *almost everywhere, and*

(iii) $\lim_{T \to \infty} \frac{1}{2T} \int_{-T}^{T} f(t) \, dt$ *exists and is equal to* $\sqrt{(\pi/2)} \mu(\{0\})$.

Proof. Since $R(\Phi)$ contains the constant functions, we can assume without loss of generality that $f \geqslant 0$. But, then, it follows from (7) that $\int \hat{h}(-u) f(u) \, du$ exists as a Lebesgue integral whenever $h \in (C_0, l)$ and $\hat{h} \geqslant 0$. Since $h(t) = e^{-|t|}$ belongs to (C_0, l) it results in particular that

$$\int \frac{f(t)}{1+t^2} \, dt < \infty \tag{11}$$

and that

$$\sqrt{(2/\pi)}\int\frac{f(t)\,dt}{1+(t-x)^2}=\int e^{-|t|-ixt}\,d\mu(t),$$

for all real x. Since

$$\int\frac{f(t)}{1+(t-x)^2}\,dt\geqslant\int_x^{x+1}\frac{f(t)\,dt}{1+(t-x)^2}\geqslant\frac{1}{2}\int_x^{x+1}f(t)\,dt$$

we see that

$$\sup_{-\infty<x<\infty}\int_x^{x+1}f(t)\,dt\leqslant\sqrt{(2\pi)}\int e^{-|t|}\,d\,|\mu|\,(t).$$

This proves part (i).

As for part (ii), if in (7) we replace $h(t)$ by $e^{-ixt}k_\lambda(t)$ we find that

$$\int\frac{1-\cos\lambda(x-u)}{\pi\lambda(x-u)^2}f(u)\,du=\frac{1}{\sqrt{(2\pi)}}\int_{|t|\leqslant\lambda}(1-|t/\lambda|)\,e^{-ixt}\,d\mu(t). \tag{12}$$

On account of (11), (ii) is now a direct consequence of the last displayed identity and Theorem 13 ([**10**], p. 28).

To prove part (iii), we put $x=0$ in (12) and replace λ by 2ε. We obtain

$$\frac{1}{\pi}\int\frac{\sin^2\varepsilon u}{\varepsilon u^2}f(u)\,du=\frac{1}{\sqrt{(2\pi)}}\int_{|t|\leqslant 2\varepsilon}(1-|t/2\varepsilon|)\,d\mu(t)$$

and, making $\varepsilon\to 0$, we deduce that

$$\lim_{\varepsilon\to 0}\frac{1}{\pi}\int\frac{\sin^2\varepsilon u}{\varepsilon u^2}f(u)\,du=\frac{\mu(\{0\})}{\sqrt{(2\pi)}}$$

which, by a well-known theorem of Wiener ([**13**], p. 139), is equivalent to (iii).

This completes the proof.

Our next task is to show that statement (ii) of Theorem 8 obtains for every f in $R(\Phi)$. To achieve this we require a lemma.

LEMMA. *Let $f\in R(\Phi)$. Then there is a positive constant M such that*

$$\left|\int k_\lambda(u)e^{iyu}f(x+u)\,du\right|\leqslant M\lambda \tag{13}$$

whenever x and y are real and $\lambda\geqslant 1$.

Proof. By hypothesis, there is a constant A such that

$$\left|\int f(x)\varphi(x)\,dx\right|\leqslant A\,\|\hat{\varphi}\|_{\infty,1}$$

for all $\varphi\in\Phi$. In particular, putting

$$\varphi(u)=e^{iyu}k_\lambda(u-x),$$

so that

$$\hat{\varphi}(t) = e^{-i(t-y)x}k_\lambda(t-y),$$

we find that

$$\left|\int \hat{k}_\lambda(u-x)e^{iyu}f(u)\,du\right| \leqslant A\,\|k_\lambda(t-y)\|_{\infty,1} \leqslant 2A\,\|k_\lambda\|_{\infty,1}, \tag{14}$$

for all real x, y. It only remains to estimate $\|k_\lambda\|_{\infty,1}$. Now

$$k_\lambda(x) = \frac{1-\cos\lambda x}{\pi\lambda x^2} = \frac{2\sin^2(\lambda x/2)}{\pi\lambda x^2},$$

and $|\sin x/x| \leqslant 2/(1+|x|)$ for all real x. Hence

$$k_\lambda(x) \leqslant \frac{\lambda}{2\pi}\frac{4}{(1+\lambda|x|/2)^2} = \frac{8\lambda}{\pi(2+\lambda|x|)^2} \leqslant \frac{8\lambda}{\pi(2+|x|)^2}$$

if $\lambda \geqslant 1$. Consequently,

$$\|k_\lambda\|_{\infty,1} \leqslant \frac{16\lambda}{\pi}\sum_{n=0}^{\infty}\frac{1}{(2+n)^2} \leqslant \frac{8\pi\lambda}{3},$$

which result, taken in conjunction with (14), gives (13) with $M = 16\lambda A/3$.

THEOREM 9. *Let $f \in R(\Phi)$ and let μ be the measure in M_∞ pertaining to f, as defined in Theorem 7. Then, for almost all real x,*

$$\sqrt{(2\pi)}f(x) = (\mathrm{C},1)\int e^{-ixt}\,d\mu(t). \tag{15}$$

Proof. According to Theorem 7, we have

$$(\mathrm{C},1)\int \hat{h}(-u)f(u)\,du = \int h\,d\mu \quad (h \in (C_0, l)).$$

In this formula, put

$$h(t) = e^{-ixt}(\hat{k}_w * k_\lambda)(t),$$

where $w, \lambda > 0$ and, as usual, k_λ denotes Fejér's kernel. Since $\hat{h}(u) = \sqrt{(2\pi)}k_w(u+x)\hat{k}_\lambda(u+x)$, we infer that

$$\sqrt{(2\pi)}\int k_w(x-u)\hat{k}_\lambda(x-u)f(u)\,du = \int (\hat{k}_w * k_\lambda)(t)\,e^{-ixt}\,d\mu(t).$$

Now, by Theorem E, $\|\hat{k}_w * k_\lambda - k_w\|_{\infty,1} \to 0$ as $\lambda \to \infty$. Hence

$$\lim_{\lambda\to\infty}\int (\hat{k}_w * k_\lambda)(t)\,e^{-ixt}\,d\mu(t) = \int \hat{k}_w(t)\,e^{-ixt}\,d\mu(t)$$

uniformly for all real x. Accordingly, to prove (15), it suffices to show that

$$\lim_{w\to\infty}\lim_{\lambda\to\infty}\sqrt{(2\pi)}\int k_w(x-u)\hat{k}_\lambda(x-u)f(u)\,du = f(x), \tag{16}$$

almost everywhere; and this we proceed to demonstrate.

Write

$$f_1(t,w,x) = \int_0^t (1-\cos wu)(f(x+u)+f(x-u))\,du,$$

and

$$f_2(t,w,x) = \int_0^t f_1(s,w,x)\,ds,$$

so that

$$f_2(t,w,x) = \int_{-t}^t (t-|u|)(1-\cos wu)f(x-u)\,du.$$

By the lemma, there is a constant B such that

$$\sup\{|f_2(t,w,x)| : w>0,\ -\infty<x<\infty\} \leqslant Bt^2 \tag{17}$$

for all $t \geqslant 1$. Now consider

$$\begin{aligned}
&\sqrt{(2\pi)}\int k_w(u)\hat{k}_\lambda(u)f(x-u)\,du \\
&\quad = \int_{-\lambda}^{\lambda}(1-|u/\lambda|)\frac{(1-\cos wu)}{\pi wu^2}f(x-u)\,du \\
&\quad = \int_0^{\lambda}(1-u/\lambda)\frac{(1-\cos wu)}{\pi wu^2}(f(x+u)+f(x-u))\,du \\
&\quad = \left(\int_0^1+\int_1^{\lambda}\right)(1-u/\lambda)\frac{(1-\cos wu)}{\pi wu^2}(f(x+u)+f(x-u))\,du \\
&\quad = I_1(w,\lambda,x)+I_2(w,\lambda,x),
\end{aligned}$$

say. We shall examine separately the behaviour of $I_1(w,\lambda,x)$ and $I_2(w,\lambda,x)$ as first λ and then w tend to infinity.

Concerning $I_2(w,\lambda,x)$ we have, on integrating by parts twice,

$$\pi wI_2(w,\lambda,x) = [g(u)f_1(u,w,x)-g'(u)f_2(u,w,x)]_{u=1}^{u=\lambda}+\int_1^{\lambda}g''(u)f_2(u,w,x)\,du,$$

where

$$g(u) = (\lambda-u)/\lambda u^2, \quad g'(u) = (u-2\lambda)/\lambda u^3,$$

and

$$g''(u) = (6\lambda-2u)/\lambda u^4.$$

Now, as $\lambda \to \infty$,

$$\int_1^\lambda g''(u) f_2(u,w,x)\,du = 6\int_1^\lambda f_2(u,w,x)\frac{du}{u^4} - \frac{2}{\lambda}\int_1^\lambda f_2(u,w,x)\frac{du}{u^3}$$

$$\to 6\int_1^\infty f_2(u,w,x)\frac{du}{u^4}$$

uniformly with respect to x and w (this is so because of (17)); and the expression in square brackets tends to

$$-f_1(1,w,x) - 2f_2(1,w,x) = \int_{-1}^1 (2|u|-3)(1-\cos wu) f(x-u)\,du.$$

Thus

$$\lim_{\lambda\to\infty} I_2(w,\lambda,x)$$

$$= \frac{1}{\pi w}\int_{-1}^1 (2|u|-3)(1-\cos wu) f(x-u)\,du + \frac{6}{\pi w}\int_1^\infty f_2(u,w,x)\frac{du}{u^4}$$

$$= O(w^{-1}) = o(1)$$

as $w \to \infty$, uniformly with respect to x on every finite interval. In other words,

$$\lim_{w\to\infty}\lim_{\lambda\to\infty} I_2(w,\lambda,x) = 0$$

uniformly on $[a,b]$, $-\infty < a < b < \infty$. Next, as $\lambda \to \infty$,

$$I_1(w,\lambda,x) = \int_{-1}^1 (1-|u/\lambda|)\frac{(1-\cos wu)}{\pi w^2 u^2} f(x-u)\,du$$

$$\to \int_{-1}^1 \frac{1-\cos wu}{\pi w u^2} f(x-u)\,du$$

by Lebesgue's dominated convergence theorem. Finally, on making $w \to \infty$, a Fejér argument shows that the last integral converges almost everywhere to $f(x)$. In other words, (16) is satisfied; and, in view of our opening remarks, it now results that

$$f(x) = \lim_{w\to\infty} \int \hat{k}_w(t)\, e^{-ixt}\, d\mu(t)$$

almost everywhere, which is equivalent to (15).

This ends the proof.

We have, however, proved a little more; and, as a bonus, we obtain the next result.

THEOREM 10. *With the hypotheses of the previous theorem,*

$$\lim_{w\to\infty}\int_a^b\left|f(x)-\int \hat{k}_w(t)\,e^{-ixt}\,d\mu(t)\right|dx=0$$

provided that $-\infty<a<b<\infty$.

Proof. Retaining the notation of the preceding theorem, we have

$$\begin{aligned}
&\hat{k}_w(t)\int e^{-ixt}\,d\mu(t)-f(x)\\
&\quad=\int_{-1}^{1}\frac{1-\cos wu}{\pi wu^2}f(x-u)\,du-f(x)+\lim_{\lambda\to\infty}I_2(w,\lambda,x)\\
&\quad=\int_{-1}^{1}\frac{1-\cos wu}{\pi wu^2}(f(x-u)-f(x))\,du-f(x)\int_{|u|>1}\frac{(1-\cos wu)}{\pi wu^2}du\\
&\qquad+\lim_{\lambda\to\infty}I_2(w,\lambda,x).
\end{aligned}$$

Taking absolute values and integrating over $[a,b]$, we obtain

$$\begin{aligned}
&\int_a^b\left|\int \hat{k}_w(t)\,e^{-ixt}\,d\mu(t)-f(x)\right|dx\\
&\quad\leqslant\int_{-1}^{1}\frac{1-\cos wu}{\pi wu^2}\int_a^b|f(x-u)-f(x)|\,dx\,du+\frac{2}{\pi w}\int_a^b|f(x)|\,dx\\
&\qquad+\int_a^b\left|\lim_{\lambda\to\infty}I_2(w,\lambda,x)\right|dx\\
&\quad=o(1)+O(w^{-1})+O(w^{-1})\\
&\quad=o(1)
\end{aligned}$$

as $w\to\infty$, since $\lim_{\lambda\to\infty}I_2(w,\lambda,x)=O(w^{-1})$ $(w\to\infty)$ uniformly with respect to x.

This is what we wanted to show.

THEOREM 11. *Let* (f_n) *be a sequence of elements in* $R(\Phi)$ *and suppose that there is a function* $f\in L_{\mathrm{loc}}$ *such that*

$$\lim_{n\to\infty}\int f_n(x)\varphi(x)\,dx=\int f(x)\varphi(x)\,dx\quad(\varphi\in C_c). \tag{18}$$

Let (μ_n) *be the sequence of measures in* M_∞ *corresponding to* (f_n) *and suppose that*

$$\sup\{\|\mu_n\|_{M_\infty}:n\geqslant 1\}<\infty. \tag{19}$$

Then $f \in R(\Phi)$ and the measure μ in M_∞ pertaining to f is the weak limit of (μ_n).*

Proof. If $\varphi \in \Phi$ then, by (7),

$$\int \varphi(x) f_n(x)\, dx = \int \hat{\varphi}\, d\mu_n \quad (n = 1, 2, \ldots).$$

Hence

$$\left| \int \varphi(x) f_n(x)\, dx \right| \leqslant \| \hat{\varphi} \|_{\infty,1} \| \mu_n \|_{M_\infty}$$

$$\leqslant \| \hat{\varphi} \|_{\infty,1} \sup_{n \geqslant 1} \| \mu_n \|_{M_\infty}$$

and, making $n \to \infty$, we deduce that

$$\left| \int f(x) \varphi(x)\, dx \right| \to \| \hat{\varphi} \|_{\infty,1} \sup_{n \geqslant 1} \| \mu_n \|_{M_\infty}$$

for all $\varphi \in \Phi$. Thus $f \in R(\Phi)$.

Let μ be the measure in M_∞ corresponding to f. On the one hand, by (7),

$$\int h\, d\mu = \lim_{\lambda \to \infty} \sqrt{(2\pi)} \int \hat{k}_\lambda(u) \hat{h}(-u) f(u)\, du$$

$$= \lim_{\lambda \to \infty} \lim_{n \to \infty} \sqrt{(2\pi)} \int \hat{k}_\lambda(u) \hat{h}(-u) f_n(u)\, du$$

$$= \lim_{\lambda \to \infty} \lim_{n \to \infty} \int k_\lambda * h\, d\mu_n$$

for all $h \in (C_0, l)$. The transition from the first to the second line is guaranteed by (18) since $\hat{k}_\lambda \hat{h} \in C_c$; and the next step is a consequence of (7) because the Fourier transform of $k_\lambda * h$ is $\sqrt{(2\pi)} \hat{k}_\lambda \hat{h}$, and $\hat{k}_\lambda$ is even.

On the other hand, if σ is a weak limiting point of (μ_n)—and there is at least one, because of condition (19)—then there is a subsequence (μ_{n_k}) such that

$$\int h\, d\sigma = \lim_{k \to \infty} \int h\, d\mu_{n_k} \quad (h \in (C_0, l)).$$

Consequently, if $h \in (C_0, l)$,

$$\int h\, d\mu = \lim_{\lambda \to \infty} \lim_{n \to \infty} \int k_\lambda * h\, d\mu_n$$

$$= \lim_{\lambda \to \infty} \int k_\lambda * h\, d\sigma$$

$$= \int h\, d\sigma$$

by Theorem E. Thus $\sigma = \mu$, and the proof is finished.

10. Applications

Theorem 11 of the last section yields the following result.

THEOREM 12. *Let* $K(x,\lambda) \to 1$ *as* $\lambda \to \infty$, *boundedly over every finite interval. Let* $f \in L_{\text{loc}}$ *and let* $\{f(x)K(x,\lambda) : \lambda > 0\}$ *be a subset of* L. *Let*

$$\sqrt{(2\pi)} f_\lambda(x) = \int e^{ixt} f(t) K(t,\lambda)\, dt.$$

Then corresponding to each of the succeeding assumptions on $\{f_\lambda : \lambda > 0\}$:

(a) $\{f_\lambda : \lambda > 0\}$ *is a bounded subset of* L^∞;

(b) $f_\lambda(x) \geqslant 0$ *for all real* x *and all* $\lambda > 0$;

(c) $\{f_\lambda : \lambda > 0\}$ *is a bounded subset of* (L, l^∞); *then* f *admits a representation*

$$\sqrt{(2\pi)} f(x) = \int e^{-ixt}\, d\mu(t) \quad \text{a.e. on } (-\infty,\infty), \tag{20}$$

where our conclusions regarding μ *are, respectively:*

(a)′ $\mu([0,x]) = \int_0^x F(s)\, ds$, *where* $F \in L^\infty$;

(b)′ μ *is a positive measure, and* $\lim_{|x|\to\infty} \mu([x,x+1]) = 0$;

(c)′ μ *belongs to* M_∞.

Proof. It is clear that (a) implies (c); and we go on to show that (b) implies (c) as well. For this purpose, put

$$h(x) = \sin^2 x/x^2.$$

Then h is even, non-negative, $h(x) \geqslant h(1) > 0$ if $|x| \leqslant 1$ and $\hat{h}$ belongs to C_c. Therefore, by Plancherel's theorem,

$$\begin{aligned} \int K(t,\lambda) f(t) \hat{h}(t) e^{ixt}\, dt &= \int h(u) f_\lambda(x+u)\, du \\ &\geqslant \int_0^1 h(u) f_\lambda(x+u)\, du \\ &\geqslant h(1) \int_x^{x+1} f_\lambda(u)\, du, \end{aligned} \tag{21}$$

because of (b) and the properties of h. As a consequence,

$$\begin{aligned} \sup_{-\infty<x<\infty} \int_x^{x+1} f_\lambda(u)\, du &\leqslant \frac{1}{h(1)} \int |K(t,\lambda) f(t) \hat{h}(t)|\, dt \\ &= O(1) \quad (\lambda \to \infty). \end{aligned}$$

Thus

$$\sup_{\lambda>0} \|f_\lambda\|_{1,\infty} < \infty,$$

and so (b) implies (c).

Suppose now that (c) obtains and put

$$\mu_\lambda(E) = \int_E f_\lambda(u)\,du,$$

so that the family of measures $(\mu_\lambda : \lambda > 0)$ is a uniformly bounded subset of M_∞. It is clear that $K(x,\lambda)f(x)$ belongs to $R(\Phi)$ for every $\lambda > 0$; and Fourier's inversion formula yields

$$\begin{aligned}\sqrt{(2\pi)}K(x,\lambda)f(x) &= (\mathrm{C},1)\int e^{-ixt}f_\lambda(t)\,dt \\ &= (\mathrm{C},1)\int e^{-ixt}\,d\mu_\lambda(t),\end{aligned}$$

almost everywhere. Since $K(x,\lambda)f(x) \to f(x)$ as $\lambda \to \infty$, boundedly over compact sets, Theorems 11 and 9 allow us to conclude that $f \in R(\Phi)$ and that it is expressible in the form (20); moreover, μ is the weak* limit of $(\mu_\lambda : \lambda > 0)$.

That (a) implies (a)′ is clear.

If f_λ obeys (b), then $(\mu_\lambda : \lambda > 0)$ is a family of positive measures, and so μ is positive. Returning to (21) and, if necessary, making λ tend to infinity through a suitable sequence of values, we infer that

$$\begin{aligned}\int f(t)\hat{h}(t)\,e^{ixt}\,dt &= \int h(t-x)\,d\mu(t) \\ &\geqslant \int_{[x,x+1]} h(t-x)\,d\mu(t) \\ &\geqslant h(1)\mu([x,x+1]),\end{aligned}$$

whence it follows from the Riemann–Lebesgue lemma that $\mu([x,x+1]) \to 0$ as $|x| \to \infty$. Thus (b) forces (20) and (b)′.

This completes the proof of the theorem.

Special cases of conditions (a) and (b) enunciated in the foregoing theorem have previously been discussed in the literature. Thus, as a special case of (a), with $K(x,\lambda) = e^{-|x|/\lambda}$, we obtain a result of Verblunsky ([**12**]) to the effect that the boundedness of the set

$$\left\{\sup_{-\infty<i<\infty}\left|\int e^{-|x|/\lambda+itx}f(x)\,dx\right| : \lambda > 0\right\}$$

is sufficient to ensure the existence of $F \in L^\infty$ such that

$$\sqrt{(2\pi)}f(x) = (\mathrm{C},1)\int e^{-ixt}F(t)\,dt \quad \text{a.e. on } (-\infty,\infty).$$

Offord ([**8**]) obtained the same conclusion by assuming the uniform boundedness of

$$\int_{-\lambda}^{\lambda}(1-|t/\lambda|)f(t)\,e^{+ixt}\,dt.$$

An extension of their results, along the lines presented here, was also indicated by González-Domínguez ([5]).

Taking $K(x,\lambda) = \hat{k}_\lambda(x)$ and applying (b), we recapture Cooper's representation theorem for generalized positive definite functions ([3]).

Assuming condition (b) with $K(x,\lambda) = e^{-|x|/\lambda}$ and assuming, in addition, that either $f(t) = O(|t|)$ $(|t| \to \infty)$ or $\lim_{T\to\infty} \int_{-T}^{T} f(t)\,dt/(1+t^2)$ exists, Verblunsky ([12]) showed that f is representable in the form

$$f(x) = \lim_{\varepsilon\to 0} \int e^{-\varepsilon|t|-ixt}\,d\rho(t),$$

where ρ is a positive measure. But, actually, as a survey of the argument used to prove Theorem 9 reveals, this result is valid without the additional requirements imposed on f by Verblunsky; furthermore, ρ is a member of M_∞.

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